Question 3

Coefficient of volume expansion: $B = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p = \frac{1}{T}$

Grashof number: $Gr = \frac{g(T_s - T_\infty)L^3}{T\nu^2}$

Question 8

$$\frac{\text{Axial conduction}}{\text{Axial convection}} = \frac{K}{\rho C_p v_z L} = \frac{1}{Pe_z}$$

plug in number $= \frac{0.6}{1000 \cdot 4200 \cdot 0.1 \cdot 1} = 1.4 \cdot 10^{-6}$

$\Rightarrow Pe_z = 1.4 \cdot 10^6 >> 1$. $\Rightarrow$ making axial conduction negligible

Question 10

$q = -K\frac{dT}{dx}$ since material II has smaller temperature drop

$\Rightarrow$ higher slope

$\Rightarrow$ higher K

10.A.5

a) $V_z = A(u^3 - u)$ $\left(u = \frac{y}{B}\right)$

$$A = \frac{(\bar{\rho} g \bar{B} \Delta T) B^2}{12\mu}$$

$$\Rightarrow V_z^{up} = A\int_{u=-1}^{u=0} [u^3 - u]\,du = \frac{A}{4} = \frac{(\bar{\rho} g \bar{B} \Delta T) B^2}{48\mu}$$

b) $T = \frac{1}{2}(T_1 + T_2) = 60°C = 332.2K$

$\bar{B} = 1/\bar{T} = 3.10^{-3} K^{-1}$

c) $(v_z^{up}) = \dfrac{980.7\,cm/s \,.\, 3.10^{-3}K^{-1} \,.\, 80C \,.\, 0.3cm^2}{48 \,.\, 0.1886\,cm^2/s}$

$= 2.3\,cm/s$

10. A.6

a) $q = -K\dfrac{dT}{dx} = -K\dfrac{(T_1 - T_2)}{x_2 - x_1} = -0.075\dfrac{(69-61)}{\frac{0.502}{12}}$

$= 14.3\, Btu/hr.ft^2$

b) $R = \dfrac{\Delta T}{q} = \dfrac{(T_2 - T_3)}{q} = \dfrac{61}{14.3} = 4.2\dfrac{F}{Btu/hr.ft^2}$

10. B1

$T(r)$

K constant

Shell energy balance

$0 = \int_S ds\,\langle q\,n\rangle$

$\Rightarrow 0 = [4\pi r^2 q_r]_r - [4\pi r^2 q_r]_{r+\Delta r}$

$\Rightarrow \frac{1}{r^2}\frac{d}{dr}(r^2 q_r) = 0 \Rightarrow \frac{d}{dr} r^2 q_r = 0 \Rightarrow r^2 q_r = C_1 \Rightarrow q_r = \frac{C_1}{r^2}$

$\frac{d}{dr} r^2\left(-K\frac{dT}{dr}\right) = 0$

$\Rightarrow \frac{d}{dr}\left(r^2\frac{dT}{dr}\right) = 0$

$$q_r = -k\frac{dT}{dr} \Rightarrow -k\frac{dT}{dr} = \frac{C_1}{r^2}$$

$$\Rightarrow -\frac{dT}{dr} = \frac{C_1}{r^2} \Rightarrow T = \frac{-C_1}{r} + C_2$$

b) $T(r=R) = T_R$

$T(r=\infty) = T_\infty$

$\Rightarrow$ plug in $\Rightarrow T_R = \frac{-C_1}{R} + C_2 \Rightarrow T_R = \frac{-C_1}{R} + T_\infty$

$\Rightarrow C_1 = (T_\infty - T_R)R$

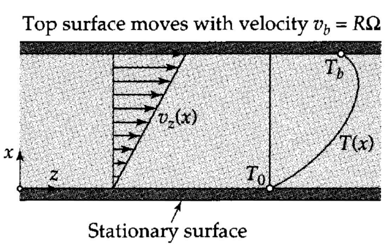
$$T_\infty = \frac{-C_1}{\infty} + C_2 \Rightarrow T_\infty = C_2$$

c) $$q_R = -k\frac{dT}{dR} = +kR(T_R - T_\infty)\frac{1}{r^2} = \frac{k(T_R - T_\infty)}{R} = h(T_R - T_\infty)$$

$$\left(\frac{hD}{k} = 2 \Rightarrow h = \frac{2k}{D} = \frac{k}{R}\right)$$

d : Bi contain k of the solid , Nu contains k of the fluid

10. B2



Shell energy balance.

$$-\int_S dS \langle q|n\rangle - \int_S dS \langle v|\tau|n\rangle + \int_S dS\rho \langle v|n\rangle = 0$$

$$\Rightarrow \{LWq_x\}_x - [LWq_x]_{x+\Delta x} + [LWV_x\tau_{xx}]_x - [LWV_x\tau_{xx}]_{x+\Delta x}$$
$$+ \cancel{[\Delta x W \rho V_z]_z - [\Delta x W \rho V_z]_{z+L}} \text{ (velocity only on z direction)}$$

$$\Rightarrow \{LWq_x\}_x - [LWq_x]_{x+\Delta x} + [LWV_x\tau_{xx}]_x - [LWV_x\tau_{xx}]_{x+\Delta x} = 0$$

$$\Rightarrow \frac{-dq_x}{dx} + \left(-\tau_{xx}\frac{dV_x}{dx}\right) = 0$$

$$\Rightarrow \frac{-dq_x}{dx} = \tau_{xx}\frac{dV_x}{dx} \Rightarrow \frac{-dq_x}{dx} = \tau_{xx}\frac{V_b}{b}$$

$$\text{As } \tau_{xx} = -u\frac{dV_x}{dx} = -u\frac{V_b}{b} \Rightarrow \frac{dq_x}{dx} = u\left(\frac{V_b}{b}\right)^2$$

$$q_x = -K\frac{dT}{dx} \Rightarrow \frac{d^2T}{dx^2} = \frac{-u}{K}\left(\frac{V_b}{b}\right)^2 \Rightarrow T = \frac{-u}{2K}\left(\frac{V_b}{b}\right)^2 x^2 + C_1x + C_2$$

$$\frac{dT}{dx} = \frac{-u}{K}\left(\frac{V_b}{b}\right)^2 x + C_1$$

BC: $T(x=0) = T_0$

$q(x=b) = 0 \Rightarrow -K\frac{dT}{dx} = 0$ when $x = b$

$$\Rightarrow T_0 = C_2$$

$$\frac{-u}{K}\left(\frac{V_b}{b}\right)^2 b + C_1 = 0 \Rightarrow C_1 = \frac{u}{K}\cdot\frac{V_b^2}{b}$$

$$\Rightarrow \text{plug in} \Rightarrow T = \frac{-u}{2K}\left(\frac{V_b}{b}\right)^2 x^2 + \frac{u}{K}\frac{V_b^2}{b}x + T_0$$

$$\Rightarrow T - T_0 = \frac{-u}{2K}\left(\frac{Vb}{b}\right)^2 x^2 + \frac{u}{K}\frac{Vb^2}{b} x$$

$$\Rightarrow \frac{T - T_0}{uVb^2/K} = -\frac{1}{2}\left(\frac{x}{b}\right)^2 + \left(\frac{x}{b}\right)$$

10.B4. $q(r)$

Shell energy balance.

$-ds\,(q\,|n) = 0$

$$\Rightarrow [2\pi r L q_r]_r - [2\pi r L q_r]_{r+\Delta r} = 0.$$

$$\Rightarrow \frac{d(r q_r)}{dr} = 0 \Rightarrow \frac{d}{dr}\left(r k \frac{dT}{dr}\right) = 0$$

$$\Rightarrow kr\frac{dT}{dr} = C_1. \Rightarrow -k\frac{dT}{dr} = \frac{C_1}{r}$$

$$K = K_0 + (K_1 - K_0)\left(\frac{T - T_0}{T_1 - T_0}\right) = K_0 + (K_1 - K_0)$$

$$\Rightarrow -(K_0 + (K_1 - K_0))\frac{dT}{dr} = \frac{C_1}{r} \Rightarrow -(T_1 - T_0)[K_0 + \tfrac{1}{2}(K_1 - K_0)] = C_1 \ln r + C_2$$

$$\Rightarrow 0 = C_1 \ln r_0 + C_2 \text{ and } -(T_1 - T_0)[K_0 + \tfrac{1}{2}(K_1 - K_0)] = C_1 \ln r_1 + C_2$$

$$\Rightarrow C_1 = \frac{-(T_1 - T_0)}{\ln(r_1/r_0)}\left[\tfrac{1}{2}(K_0 + K_1)\right] \Rightarrow Q = 2\pi r_0 L q_r(r_0) = 2\pi L \frac{(T_0 - T_1)}{\ln(r_1/r_0)}\left[\tfrac{1}{2}(K_0 + K_1)\right]$$

b) $r_1/r_0 = 1 + \varepsilon \Rightarrow \varepsilon = (r_1/r_0) - 1 = (r_1 - r_0)/r_0$

Substitute into the expression for Q we get

$$Q = 2\pi L r_0 \left[\tfrac{1}{2}(K_0 + K_1)\right]\frac{T_0 - T_1}{r_1 - r_0}$$

10.B7 Energy shell balance. (v_z), $q_z = 0$

$$0 = -\int_S ds\, \rho\left(\cancel{\frac{1}{2}v^2} + \cancel{\phi} + u\right)(v|n) - \int_S ds(q|n) - \cancel{\int_S ds\langle v|\pi|n\rangle} + \cancel{\int_V dV}$$

$$\Rightarrow 0 = -\int_S ds\, \rho u\langle v|n) - \int_S ds(q|n)$$

$$= [2BW\Delta x\, \rho u V_z]_z - [2BW\Delta x\, \rho u V_z]_{z+\Delta z} + \cancel{[2BW\Delta z q_z]_z}$$
$$\cancel{-[2BW\Delta z q_z]_{z+\Delta z}} + [2BW\Delta z\, q_x]_x - (2BW\Delta z q_x]_{x+\Delta x}$$

$$\Rightarrow [2BW\Delta x\, \rho u V_z]_z - [2BW\Delta x\, \rho u V_z]_{z+\Delta z} + [2BW\Delta x\, q_x]_x -$$
$$[2BW\Delta x\, q_x]_{x+\Delta x} = 0$$

$$\Rightarrow \rho V_z \frac{\partial u}{\partial z} = \frac{-\partial q_x}{\partial x}$$

$$\Rightarrow \rho V_z C_p \frac{\partial T}{\partial z} = \frac{-\partial q_x}{\partial x}$$

$$\Rightarrow \rho V_{z,max}\left[1-\left(\frac{x}{B}\right)^2\right] C_p \frac{\partial T}{\partial z} = \frac{-\partial\left(-k\frac{dT}{dx}\right)}{\partial x}$$

$$\Rightarrow \rho V_{z,max}\left[1-\left(\frac{x}{B}\right)^2\right] \bar{C_p} \frac{\partial T}{\partial z} = k\frac{d^2T}{\partial x^2}$$

b)
$$\left[1-\left(\frac{x}{B}\right)^2\right]\frac{\partial T}{\partial z} = \frac{K}{\rho V_{z,max}.C_p}\frac{\partial T^2}{\partial x^2}$$

$$\Rightarrow \left[1-\left(\frac{x}{B}\right)^2\right]\frac{\partial T}{\partial z} = \frac{\partial f}{\partial z}\cdot\frac{\partial T^2}{\partial x^2}$$

$$\Rightarrow \left[1-\left(\frac{x}{B}\right)^2\right]\frac{\partial T}{\partial z}\cdot\frac{\partial z}{\partial f} = \frac{\partial T}{\partial x} \qquad \Rightarrow \left[1-\left(\frac{x}{B}\right)^2\right]\cdot\frac{\partial T}{\partial f} = \frac{\partial T}{\partial x}$$

since q_0 is constant, B constant and K is constant $\Rightarrow \left[1-(\sigma)^2\right]\frac{\partial\theta}{\partial f} = \frac{\partial^2\theta}{\partial(\sigma)}$

c) $\theta(\sigma, \xi) = C_0 \xi + \Psi(\sigma)$

$$\Rightarrow \frac{\partial^2 \Psi}{\partial \sigma^2} = C_0(1-\sigma^2) \Rightarrow \Psi(\sigma) = C_0\left(\frac{1}{2}\sigma^2 - \frac{1}{12}\sigma^4\right) + C_1\sigma + C_2$$

$$\Rightarrow \theta(\sigma, \xi) = C_0 \xi + C_0\left(\frac{1}{2}\sigma^2 - \frac{1}{12}\sigma^4\right) + C_1\sigma + C_2$$

At $\sigma = \pm 1 \Rightarrow C_1 = 0$, $C_0 = \frac{3}{2}$

$$\xi = \int_0^1 \theta(\sigma, \xi)(1-\sigma^2)\,d\sigma \Rightarrow C_2 = \frac{-39}{280}$$

$$\Rightarrow \theta(\sigma, \xi) = \frac{3}{2}\xi + \frac{3}{2}\left(\frac{1}{2}\sigma^2 - \frac{1}{12}\sigma^4\right) - \frac{39}{280}$$

10 D1

Shell energy balance

$$[2\pi 2B r q_r]_r - [2\pi 2B r q_r]_{r+\Delta r} + 4\pi r \Delta r h (T - T_a) = 0$$

$$\Rightarrow -\frac{d}{dr} r q_r - \frac{rh}{B}(T - T_a) = 0$$

$$\Rightarrow \frac{d}{dr}\left(rk\frac{dT}{dr}\right) = \frac{rh}{B}(T - T_a)$$

$$\Rightarrow \frac{dT^2}{dr^2} = \frac{rh}{Bk}(T - T_a)$$

BC: $T(r = R_0) = T_0$

$$r = R_1 \Rightarrow q_x(2B) = 0 \Rightarrow \frac{dT}{dr}(2B) = 0$$

$\xi = r/R_0 \quad , \quad \theta = \dfrac{T - T_a}{T_0 - T_a}$

$$\Rightarrow \frac{1}{\xi}\frac{d}{d\xi}\left(\xi \frac{d\theta}{d\xi}\right) - \left(\frac{hR_0^2}{BK}\right)\theta = 0$$

$$\Rightarrow \theta(\xi) = C_1 I_0(B\xi) + C_2 K_0(B\xi)$$

BC: $\theta(\xi = 1) = 1 \quad \Rightarrow \quad 1 = C_1 I_0(\beta) + C_2 K_0(B)$

$\left.\dfrac{d\theta}{d\xi}\right|_{\xi = \frac{R_1}{R_0}} = 0 \qquad 0 = C_1 B I_1(BR_1/R_0) - C_2 BK_1(BR_1 - R_0)$

$$\Rightarrow C_1 = \frac{BK_1(BR_1/R_0)}{I_0(B)BK_1(BR_1/R_0) + \beta I_1(BR_1/R_0)K_0(\beta)}$$

$$C_2 = \frac{BI_1(BR_1/R_0)}{I_0(B)(BK_1)(BR_1/R_0) + BI_1(BR_1/R_0)K_0(B)}$$

$$\Rightarrow \theta(\xi) = \frac{I_0(B\xi)K_1(BR_1/R_0) + I_1(BR_1/R_0)K_0(B\xi)}{I_0(\beta)K_1(BR_1/R_0) + I_1(BR_1/R_0)K_0(\beta)}$$

b) Heat loss

$$Q = 2\pi R_0(2B)k\left.\frac{dT}{dr}\right|_{r=R_0} = 4\pi BK(T_0 - T_a)\left.\frac{d\theta}{d\xi}\right|_{\xi_1}$$

$$= 4\pi BK\beta(T_0 - T_a)\left[\frac{I_1(BR_1/R_0)K_1(\beta) - I_1(\beta)K_1(BR_1/R_0)}{I_1(BR_1/R_0)K_0(\beta) + I_0(\beta)K_1(BR_1/R_0)}\right]$$